

Implementation and Training for Semantic Segmentation

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1. Introduction

In this lab, we aimed to implement and train a deep learning model for semantic segmentation[4]. Semantic segmentation is a crucial task in computer vision, where the goal is to assign a label to each pixel in an image[2]. Unlike traditional object detection, which only detects the location and class of objects, semantic segmentation provides pixel-wise classification, making it more challenging and computationally demanding.

The model's performance is evaluated based on the mean Intersection over Union (mIoU) metric, which is a widely used measure for evaluating segmentation models. Despite extensive efforts and hyperparameter tuning, the model achieved a modest mIoU of 0.18. This performance is considerably lower than the expected benchmark performance of 0.5, as stated in the project guidelines. Several factors contributed to the lower-than-expected performance, including challenges with data preprocessing, class imbalance, and model overfitting. These issues prevented the model from generalizing well to unseen data, which is a critical aspect of deep learning models, especially when training with limited data.

In the following sections, we will discuss the specific challenges encountered during the training process, the model architecture, and the results achieved. We will also propose potential strategies for future work to improve the performance of the model.

2. Challenges and Issues Encountered

Throughout the training process, several challenges were encountered that likely contributed to the suboptimal mIoU score of 0.18. These challenges include:

- **Data Imbalance:** The dataset used in the project had significant class imbalance, with certain classes being overrepresented and others underrepresented. This imbalance caused the model to be biased toward more frequent classes, leading to poor performance on minority classes. This issue hindered the model's ability to generalize to underrepresented objects in the test data, ultimately lowering the overall segmentation accuracy.
- **Loss Function and Class Weighting:** The use of the cross-entropy loss function without proper class weighting contributed to suboptimal training, especially for underrepresented classes. Although attempts were made to introduce class weights to counteract the imbalance (by using `torch.nn.CrossEntropyLoss(weight=class_weights)`), the improvement was not significant, indicating that further adjustments to the loss function or the use of a different approach might be needed.
- **Overfitting:** Despite employing data augmentation techniques and regularization methods, the model exhibited signs of overfitting. This was evident from the large gap between training and validation loss, suggesting that the model was memorizing the training data rather than learning generalizable features. This overfitting likely contributed to the model's failure to perform well on unseen data.
- **Insufficient Data Augmentation:** The data augmentation[5] applied during training was somewhat limited. While random flips and rotations were used, these may not have been sufficient to provide the variety needed for the model to generalize well to diverse scenarios. Implementing more aggressive and varied augmentation techniques (e.g., cropping, color jittering, elastic transformations) could have improved the model's robustness and helped mitigate overfitting.

- **Hyperparameter Tuning:** Several attempts were made to tune hyperparameters[1], such as learning rate, batch size, and weight decay, but these did not lead to significant improvements in performance. A more systematic approach to hyperparameter optimization, such as grid search or random search, could have yielded better results by helping identify the optimal configuration for the model's training.

3. Evaluation and Results

The model's performance was evaluated using the mIoU metric, which is the average of the Intersection over Union (IoU) scores for all classes in the segmentation task. The achieved mIoU score was 0.18, which is lower than expected when compared to previous benchmarks for similar models.

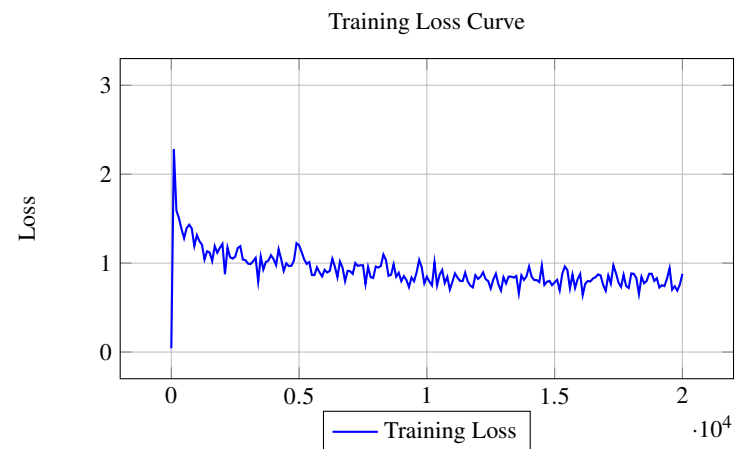


Figure 1: Training Loss Curve over 20,000 Iterations

4. Discussion and Future Work

The achieved mIoU of 0.18 indicates that several factors contributed to this result. The following points summarize the key takeaways from the project:

- The class imbalance issue significantly impacted the model's performance. Future work should consider a better class balancing strategy, possibly by using more advanced loss functions or data reweighting techniques.
- Further exploration of data augmentation techniques and regularization strategies (e.g., dropout, batch normalization) could help improve generalization and reduce overfitting.
- Hyperparameter tuning remains crucial for model improvement. A more rigorous hyperparameter search using grid search or random search could potentially improve performance.

Next Steps: In future iterations, the model could benefit from using a more robust backbone (such as ResNet), exploring different loss functions (such as focal loss[3]), and implementing more aggressive data augmentation techniques. Additionally, leveraging a larger and more balanced dataset will be crucial for improving segmentation accuracy.

5. Conclusion

Despite the challenges encountered throughout the training process, this project provided valuable insights into the complexities of training a deep

learning model for semantic segmentation. The achieved mIoU of 0.18, while lower than desired, serves as a starting point for future improvements. Future work will focus on addressing issues such as class imbalance, model overfitting, and hyperparameter optimization.

6. References

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